ELECTROCHEMISTRY

Points to be discussed

Electrochemistry
Electric conductor
Electrolysis and Electrolytic cell
Mechanism of electrolytic cell
Electrochemical series
Application of electrolysis
Electroplating
Electrochemical cell
Voltaic cell
Difference between Electrolytic cell and Electrochemical cell
Types of Battery
Cell potential

Electrochemistry

The branch of Physical Chemistry which deals with the relationship between electrical energy and chemical energy and their inter conversion

Electric conductor

a. Metallic conductor or electronic conductor
b. Electrolytic conductor

a. Metallic conductor

Metals pass electricity with the help of moving electrons

b. Electrolytic conductor

Conduct electric current either by aqueous solutions or in the molten state

Types of Electrolytic Conductor

Electrolytic conductor

Strong electrolyte
e. g.; Strong acid, strong alkali

Weak electrolyte
e. g.; weak acid, weak alkali

Electrolysis

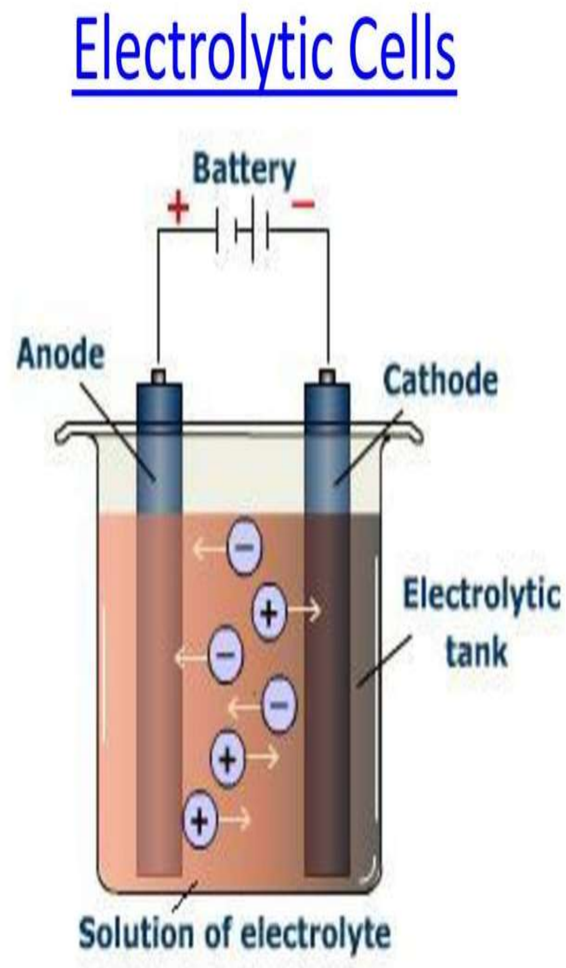
The process of chemical decomposition of an electrolyte in the molten state or in an aqueous solution by passing electric current through it is called electrolysis.

In electrolysis, electrical energy is convert in to chemical energy.

Electrolytic Cells

- An **electrolytic cell** is an electrochemical cell that drives a non-spontaneous redox reaction in the presence of **electric current supplied by an external source**.

- They are often used to electrolysis process.



Terms in electrolysis

Electrode is a rod of metal or graphite through which an electric current flows into or out of an electrolytic solution

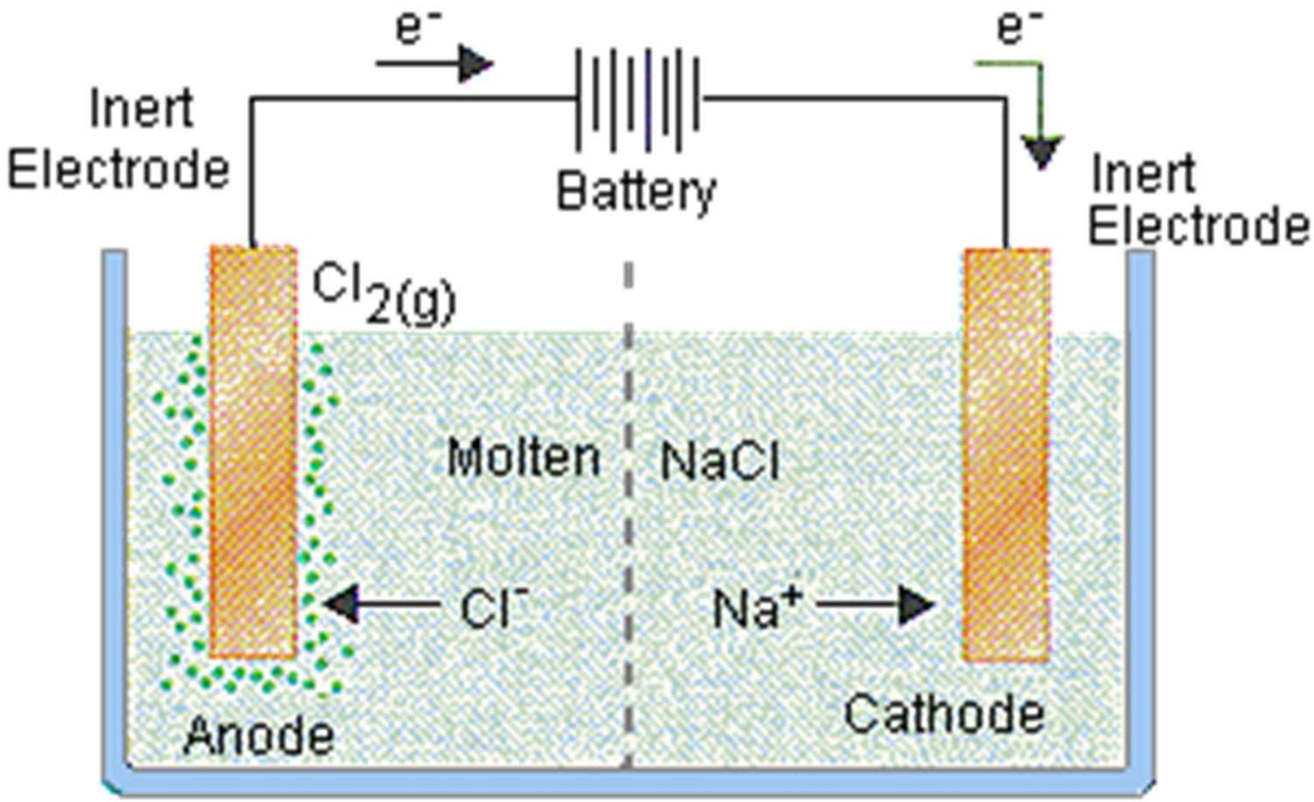
Anode is the positive electrode of an electrolytic cell

Anion is a negatively charged ion which is attracted to the anode

Cathode is the negative electrode of an electrolytic cell

Cation is a positively charged ion which is attracted to the cathode

Mechanism of electrolysis



Anode reaction

$$Cl^- = \frac{1}{2}Cl_2 + e^-$$

Cathode reaction

$$Na^+ + e \rightarrow Na$$

Electrochemical series

Cations	Anions
K^+	SO_4^{2-}
Ca^{2+}	
Na^+	NO_3^-
Mg^{2+}	
Al^{3+}	Cl^-
Zn^{2+}	
Fe^{2+}	Br^-
Pb^{2+}	
H^+	I^-
Cu^{2+}	
Hg^{2+}	OH^-
Ag^{2+}	
Au^{3+}	
Pt^{4+}	
Reducing ability of cation increases ↓	Oxidative ability of anion increases ↓

What is the electrochemical series?

This is a list of elements in order of their ability to be reduced.

For cations, the higher the element in the series, the less likely it is that this will gain electrons (that is be reduced).

For anions, the higher it is on the series the less likely will it lose electrons (that is be oxidized)

Application of Electrolysis

1. Electroplating
2. Extraction of metals
3. Purification of metals
4. Manufacture of chemicals

Principle of Electroplating

- In electroplating, **the anode is made up of the metal which we want to use to coat the surface of another metal.**
- A salt solution containing the metal that makes up the anode is used. During **electrolysis**, the anode metal is oxidized and goes into solution as positive ions.
- **These positive ions are then reduced on the surface of the cathode** (the metal we are coating)

Electroplating

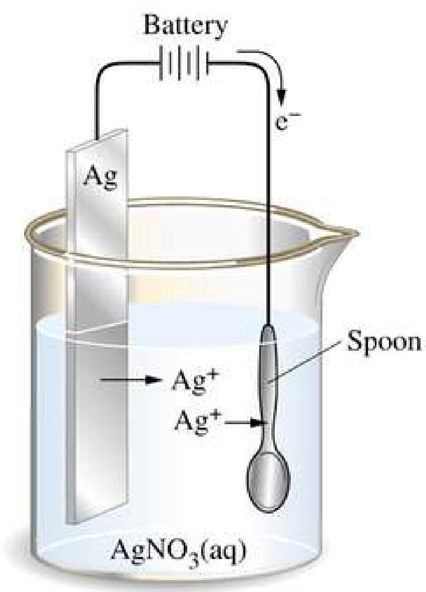


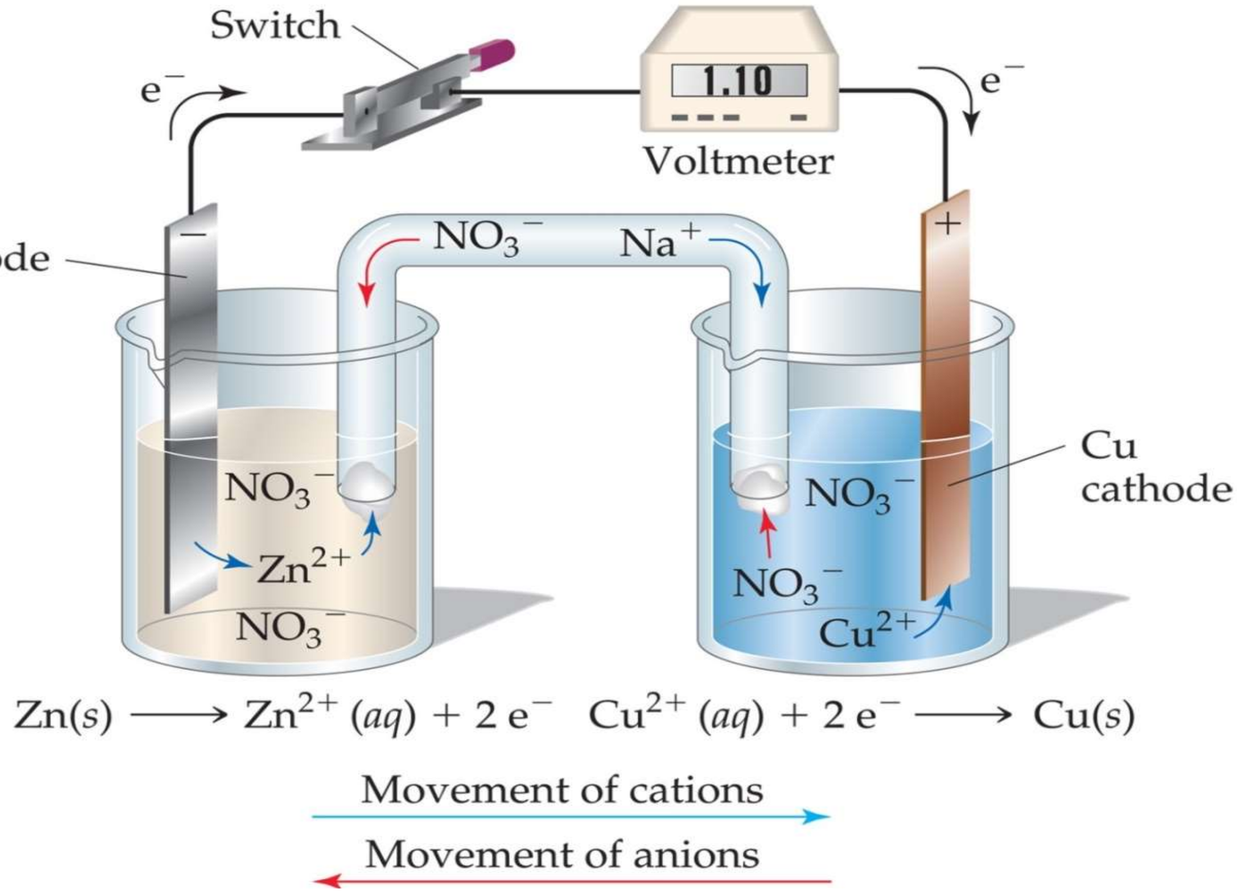
Fig.: Electroplating on a spoon

Electrochemical Cell (Battery)

What is electrochemical cell?

- **An instrumental arrangement of current flow where chemical reaction results In the production of electric current.**
- **We call such a setup :**
  - **Voltaic cell**
  - **Daniel cell**
  - **Battery**
  - **Electrochemical cell**
  - **Galvanic cell**

Typical Voltaic Cells



- The **oxidation** occurs at the **anode.**
- The **reduction** occurs at the **cathode.**
- **Electrons** flow through the **wire!**

Difference between electrolytic cell and Electrochemical cell

Electrolytic cell	Electrochemical cell
1. Electricity is passed from external source	1. Electricity is produced by chemical reactions
2. Electrical energy is converted in to chemical energy	2. chemical energy is converted in to Electrical energy
3. Metal ions cause the flow of electricity	3. Free electron cause the flow of electricity
4. Anode is positively and Cathode negatively charged	4. Anode is negatively and Cathode positively charged
5. Redox reaction is non-spontaneous	5. Redox reaction is spontaneous

Types of Battery

They are two types of batteries-

▪**Primary batteries**

•**Secondary or storage batteries**

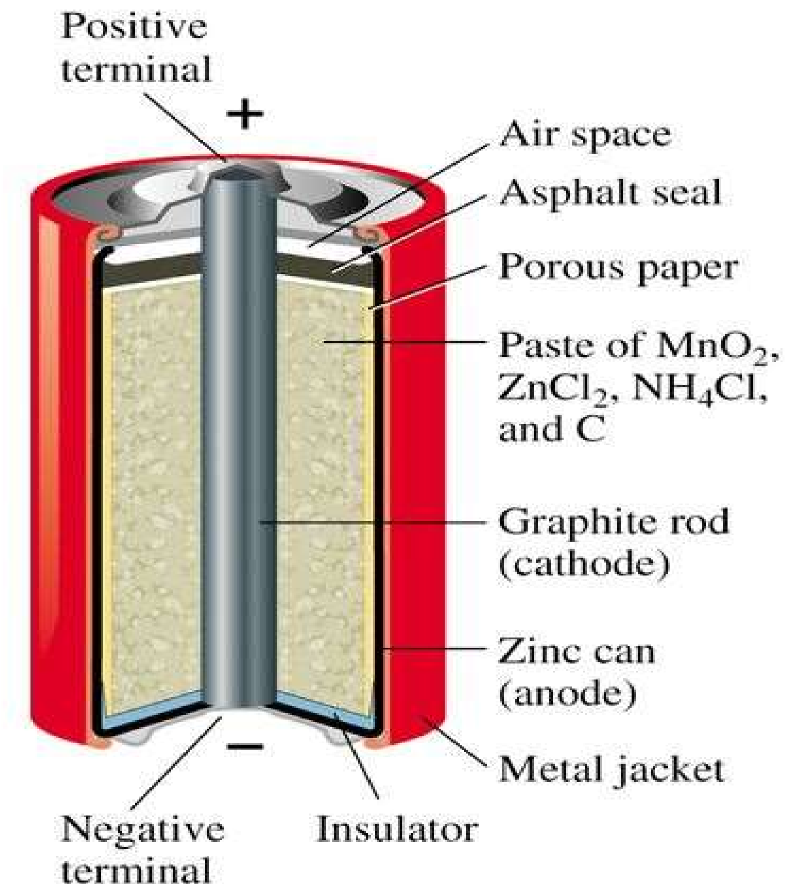
Primary batteries

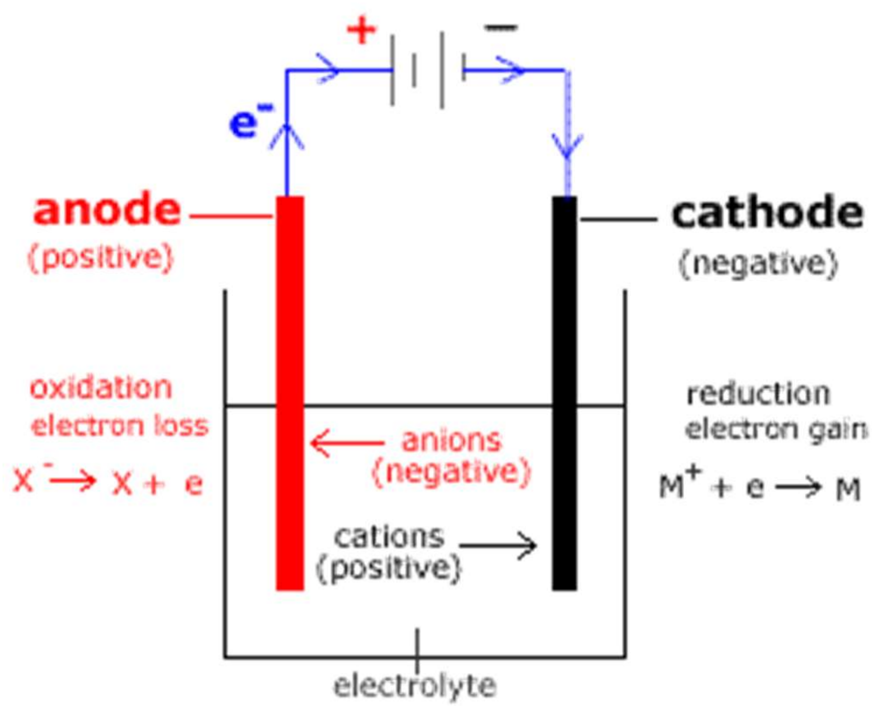
- **one continuous discharge,**
- **cannot be reused and re-charged.**
- **Use in flashlights.**

A secondary or storage battery

- **change is reversible.**
- **After the battery has discharged, it is brought back to a charged state.**
- **lead- (sulfuric) acid battery.**

Primary Battery

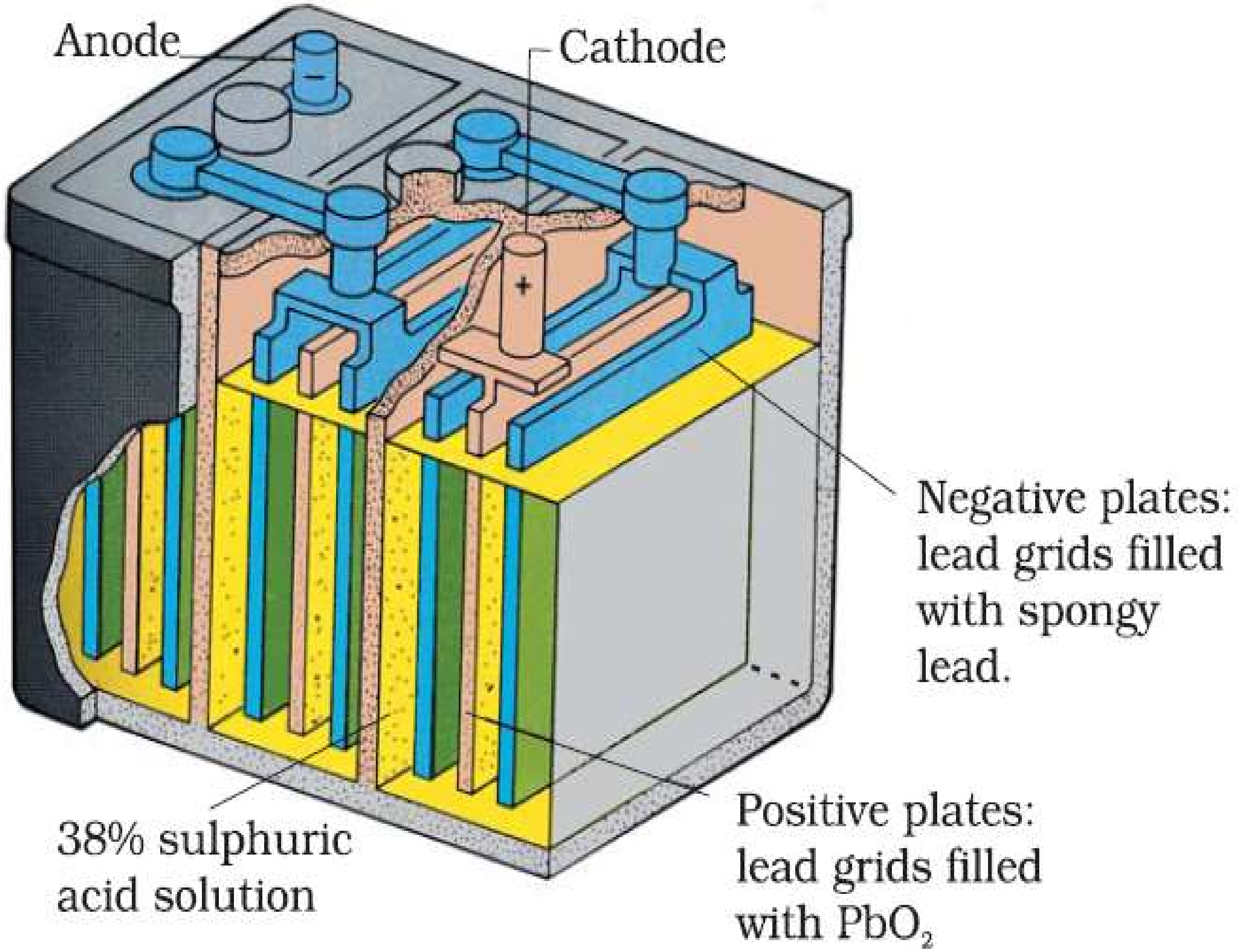




Discharge reactions in a primary battery:

- Anode: $Zn \rightarrow Zn^{2+} + 2e^{-}$
- Cathode: $2NH_4^{+} + 2MnO_2 + 2e^{-} \rightarrow Mn_2O_3 + H_2O + 2NH_3$
- However, this is complicated by the fact that the ammonium ion produces 2 gaseous products:
- $2NH^{4+} + 2e^{-} \rightarrow 2NH_3 + H_2$
- These products must be absorbed in order to prevent build up of pressure in the vessel. This occurs by the mechanism:
- $ZnCl_2 + 2NH_3 \rightarrow Zn(NH_3)_2Cl_2$
- $2MnO_2 + H_2 \rightarrow Mn_2O_3 + H_2O$

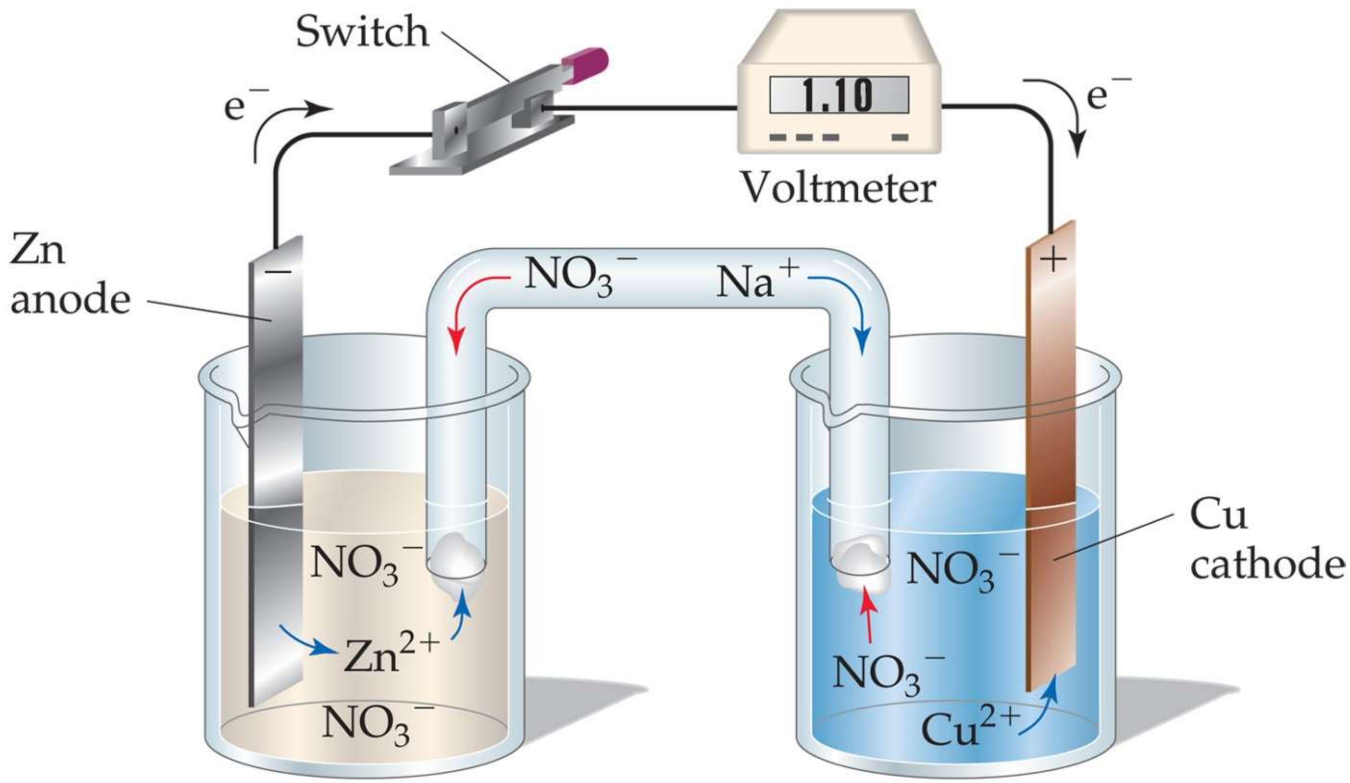
Secondary Battery



Discharge and charge reactions in a secondary battery:

- **Anode (Oxidation)**:
- $Pb\ (s) + H_2SO_4\ (aq) \rightarrow PbSO_4\ (s) + 2e^- + 2H^+\ (aq)$
- **Cathode (Reduction)**:
- $PbO_2\ (s) + 2H^+ + H_2SO_4\ (aq) + 2e^- \rightarrow PbSO_4\ (s) + 2H_2O\ (l)$
- **Overall Reaction**:
- $\rightarrow$ (discharging)
- $PbO_2\ (s) + Pb\ (s) + 2H_2SO_4\ (aq) = 2PbSO_4\ (s) + 2H_2O\ (l)$
- $\leftarrow$ (charging)
- Repeated charging causes hydrolysis of H_2O into H_2 and O_2, so distilled water is added.

Cell Potentials



$Zn(s) \longrightarrow Zn^{2+}(aq) + 2\,e^-$ $\quad Cu^{2+}(aq) + 2\,e^- \longrightarrow Cu(s)$

E_{an} = +0.76 V

E_{cat} = +0.34 V

E_{cell}= +0.34 V +0.76 V

= +1.10 V

Thank you very much for your time